

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium (LABORATORY PERFORMANCE)

COURSE CODE: CSA09

**COURSE NAME: Programming in
Java**

COURSE OUTCOME COVERED

CO4: *Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 1

Total Marks: 50

ANNEXURE A

LABORATORY PERFORMANCE

Code of Conduct:

*I Prethivika.G(192521085) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:Prethivika G

ANNEXURE

LABORATORY PERFORMANCE

1. Student Registration Applet

Develop a Java Applet for a university student registration form. The applet should display labels and input controls for Student Name, Register Number, Department, Gender, and Email ID. The application should use appropriate **AWT controls** such as Label, TextField, Checkbox, CheckboxGroup, and Button. Use a suitable **layout manager** to organize the controls. When the student clicks the **Submit** button, the entered information should be displayed within the applet window.

The student must demonstrate:

1. Applet initialization and architecture.
2. AWT controls.
3. Layout management.
4. Button event handling.
5. Appropriate event listener interface.
6. Display of user-entered information.

2. Simple Calculator Applet

Develop an Applet-based calculator that accepts two numbers and provides buttons for **Addition, Subtraction, Multiplication, and Division**.

The application should use TextField, Label, and Button controls. Appropriate event listeners must be implemented so that clicking each button performs the corresponding operation.

The program must handle invalid numeric input and division by zero appropriately.

The student must demonstrate:

1. Basic Applet structure.
2. AWT controls.
3. Event sources and event objects.
4. ActionListener.
5. Exception handling.
6. Display methods.

3. Course Selection Applet

Develop a university course selection Applet where students can select:

- Department
- Semester
- Course

- Elective type

Use Choice, List, Checkbox, and Button controls wherever appropriate. When the student clicks **Register**, display the selected courses and other information. Use appropriate layout managers and event listeners.

4. Traffic Signal Control Applet

Develop an AWT Applet that simulates a traffic signal with three options:

- Red
- Yellow
- Green

Use appropriate controls to allow the user to select a signal. When the user selects a signal, display the corresponding message such as **STOP**, **WAIT**, or **GO**.

Use appropriate event sources, event classes, and listener interfaces.

Develop a Java Applet for a university student registration form. The applet should display labels and input controls for Student Name, Register Number, Department, Gender, and Email ID. The application should use appropriate AWT controls such as Label, TextField, Checkbox, CheckboxGroup, and Button. Use a suitable layout manager to organize the controls. When the student clicks the Submit button, the entered information should be displayed within the applet window.

The student must demonstrate:

1. Applet initialization and architecture.
2. AWT controls.
3. Layout management.
4. Button event handling.
5. Appropriate event listener interface.
6. Display of user-entered information.

PROGRAM:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;
/*
<applet code="StudentRegistrationApplet.class" width="750" height="600">
</applet>
*/
public class StudentRegistrationApplet extends Applet implements ActionListener {
    // AWT Controls
    Label title, nameLabel, regLabel, deptLabel, genderLabel, emailLabel, resultLabel;
    TextField nameField, regField, emailField;
    Choice deptChoice;
    Checkbox male, female, other;
    CheckboxGroup genderGroup;
    Button submitButton, clearButton;
    // Applet initialization
    public void init() {

        // Set background and layout
        setBackground(new Color(235, 245, 255));
        setLayout(new BorderLayout(15, 15));
        // ----- HEADER -----
        Panel header = new Panel();
        header.setBackground(new Color(45, 85, 150));
        title = new Label("STUDENT REGISTRATION FORM", Label.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 26));
        title.setForeground(Color.WHITE);
        header.add(title);
        add(header, BorderLayout.NORTH);
        // ----- FORM PANEL -----
        Panel form = new Panel(new GridLayout(5, 2, 15, 15));
        form.setBackground(new Color(220, 235, 250));
        // Student Name
```

```

nameLabel = new Label("Student Name:");
nameLabel.setFont(new Font("Arial", Font.BOLD, 18));
nameField = new TextField();
nameField.setFont(new Font("Arial", Font.PLAIN, 18));
// Register Number
regLabel = new Label("Register Number:");
regLabel.setFont(new Font("Arial", Font.BOLD, 18));
regField = new TextField();
regField.setFont(new Font("Arial", Font.PLAIN, 18));
// Department
deptLabel = new Label("Department:");
deptLabel.setFont(new Font("Arial", Font.BOLD, 18));

deptChoice = new Choice();
deptChoice.add("Information Technology");
deptChoice.add("Computer Science");
deptChoice.add("Electronics");
deptChoice.add("Mechanical");
deptChoice.add("Civil");
deptChoice.setFont(new Font("Arial", Font.PLAIN, 17));
// Gender
genderLabel = new Label("Gender:");
genderLabel.setFont(new Font("Arial", Font.BOLD, 18));
Panel genderPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
genderGroup = new CheckboxGroup();
male = new Checkbox("Male", genderGroup, true);
female = new Checkbox("Female", genderGroup, false);
other = new Checkbox("Other", genderGroup, false);
male.setFont(new Font("Arial", Font.PLAIN, 17));
female.setFont(new Font("Arial", Font.PLAIN, 17));
other.setFont(new Font("Arial", Font.PLAIN, 17));
genderPanel.add(male);
genderPanel.add(female);
genderPanel.add(other);
// Email
emailLabel = new Label("Email ID:");
emailLabel.setFont(new Font("Arial", Font.BOLD, 18));
emailField = new TextField();
emailField.setFont(new Font("Arial", Font.PLAIN, 18));

// Add controls to form
form.add(nameLabel);
form.add(nameField);
form.add(regLabel);
form.add(regField);
form.add(deptLabel);
form.add(deptChoice);
form.add(genderLabel);

```

```

form.add(genderPanel);
form.add(emailLabel);
form.add(emailField);
add(form, BorderLayout.CENTER);
// ----- BUTTON PANEL -----
Panel buttonPanel = new Panel(new FlowLayout());
submitButton = new Button("SUBMIT");
clearButton = new Button("CLEAR");
submitButton.setFont(new Font("Arial", Font.BOLD, 18));
clearButton.setFont(new Font("Arial", Font.BOLD, 18));
submitButton.setBackground(new Color(40, 180, 100));
submitButton.setForeground(Color.WHITE);
clearButton.setBackground(new Color(220, 70, 70));
clearButton.setForeground(Color.WHITE);
// Event handling
submitButton.addActionListener(this);
clearButton.addActionListener(this);
buttonPanel.add(submitButton);
buttonPanel.add(clearButton);
// Result area
resultLabel = new Label("Enter details and click SUBMIT", Label.CENTER);
resultLabel.setFont(new Font("Arial", Font.BOLD, 17));
resultLabel.setForeground(new Color(40, 70, 130));
Panel bottomPanel = new Panel(new BorderLayout());
bottomPanel.setBackground(new Color(210, 225, 245));
bottomPanel.add(buttonPanel, BorderLayout.NORTH);
bottomPanel.add(resultLabel, BorderLayout.SOUTH)
add(bottomPanel, BorderLayout.SOUTH);
}
// Button event handling
public void actionPerformed(ActionEvent e) {
    // SUBMIT button
    if (e.getSource() == submitButton) {
        String name = nameField.getText();
        String reg = regField.getText();
        String dept = deptChoice.getSelectedItem();
        String gender = genderGroup.getSelectedCheckbox().getLabel();
        String email = emailField.getText();
        // Check empty fields
        if (name.equals("") || reg.equals("") || email.equals("")) {
            resultLabel.setText("Please fill all the fields!");
            resultLabel.setForeground(Color.RED);
            return;
        }
        // Display entered information
        resultLabel.setText(
            "Registered: " + name +
            " | Reg No: " + reg +

```

```

        " | Dept: " + dept +
        " | Gender: " + gender +
        " | Email: " + email
    );
    resultLabel.setForeground(new Color(20, 130, 70));
}
// CLEAR button
if (e.getSource() == clearButton) {
    nameField.setText("");
    regField.setText("");
    emailField.setText("");
    deptChoice.select(0);
    genderGroup.setSelectedCheckbox(male);
    resultLabel.setText("Form cleared. Enter new details.");
    resultLabel.setForeground(new Color(40, 70, 130));
}
}
}

```

OUTPUT:

Applet Viewer: StudentRegistrationApplet.class

Applet

STUDENT REGISTRATION FORM

Student Name: billy

Register Number: 1003

Department: Electronics

Gender: ☒ Male ☐ Female ☐ Other

Email ID: billy1003@gmail.com

Registered: billy | Reg No: 1003 | Dept: Electronics | Gender: Male | Email: billy1003@gmail.com

2. Simple Calculator Applet

Develop an Applet-based calculator that accepts two numbers and provides buttons for Addition, Subtraction, Multiplication, and Division. The application should use TextField, Label, and Button controls. Appropriate event listeners must be implemented so that clicking each button performs the corresponding operation.

The program must handle invalid numeric input and division by zero appropriately.

The student must demonstrate:

1. Basic Applet structure.
2. AWT controls.
3. Event sources and event objects.
4. ActionListener.
5. Exception handling.
6. Display methods.

PROGRAM:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;
/*
<applet code="CalculatorApplet.class" width="700" height="500">
</applet>
*/
public class CalculatorApplet extends Applet implements ActionListener {
    Label title, num1Label, num2Label, resultLabel;
    TextField num1Field, num2Field;
    Button addButton, subButton, mulButton, divButton, clearButton;
    public void init() {
        setBackground(new Color(235, 245, 255));
        setLayout(new BorderLayout(15, 15));
        Panel header = new Panel();
        header.setBackground(new Color(45, 85, 150));
        title = new Label("SIMPLE CALCULATOR", Label.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 28));
        title.setForeground(Color.WHITE);
        header.add(title);
        Panel mainPanel = new Panel(new BorderLayout(10, 10));
        Panel inputPanel = new Panel(new GridLayout(2, 2, 15, 15));
        inputPanel.setBackground(new Color(220, 235, 250));
        num1Label = new Label("First Number:");
        num1Label.setFont(new Font("Arial", Font.BOLD, 19));
        num2Label = new Label("Second Number:");
        num2Label.setFont(new Font("Arial", Font.BOLD, 19));
        num1Field = new TextField();
        num2Field = new TextField();
        num1Field.setFont(new Font("Arial", Font.PLAIN, 19));
        num2Field.setFont(new Font("Arial", Font.PLAIN, 19));
        inputPanel.add(num1Label);
        inputPanel.add(num1Field);
```

```

        inputPanel.add(num2Label);
        inputPanel.add(num2Field);
        mainPanel.add(inputPanel, BorderLayout.CENTER);
        resultLabel = new Label("Enter two numbers and select an operation.",
Label.CENTER);
        resultLabel.setFont(new Font("Arial", Font.BOLD, 20));
        resultLabel.setForeground(new Color(40, 70, 130));
        mainPanel.add(resultLabel, BorderLayout.SOUTH);
        add(header, BorderLayout.NORTH);
        add(mainPanel, BorderLayout.CENTER);
        Panel buttonPanel = new Panel(new GridLayout(2, 3, 12, 12));
        buttonPanel.setBackground(new Color(210, 225, 245));
        addButton = new Button("ADD");
        subButton = new Button("SUBTRACT");
        mulButton = new Button("MULTIPLY");
        divButton = new Button("DIVIDE");
        clearButton = new Button("CLEAR");
        addButton.setFont(new Font("Arial", Font.BOLD, 18));
        subButton.setFont(new Font("Arial", Font.BOLD, 18));
        mulButton.setFont(new Font("Arial", Font.BOLD, 18));
        divButton.setFont(new Font("Arial", Font.BOLD, 18));
        clearButton.setFont(new Font("Arial", Font.BOLD, 18));
        addButton.setBackground(new Color(40, 180, 100));
        addButton.setForeground(Color.WHITE);
        subButton.setBackground(new Color(230, 150, 50));
        subButton.setForeground(Color.WHITE);
        mulButton.setBackground(new Color(80, 130, 210));
        mulButton.setForeground(Color.WHITE);
        divButton.setBackground(new Color(150, 80, 190));
        divButton.setForeground(Color.WHITE);
        clearButton.setBackground(new Color(220, 70, 70));
        clearButton.setForeground(Color.WHITE);
        addButton.addActionListener(this);
        subButton.addActionListener(this);
        mulButton.addActionListener(this);
        divButton.addActionListener(this);
        clearButton.addActionListener(this);
        buttonPanel.add(addButton);
        buttonPanel.add(subButton);
        buttonPanel.add(mulButton);
        buttonPanel.add(divButton);
        buttonPanel.add(clearButton);
        add(buttonPanel, BorderLayout.SOUTH);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == clearButton) {
            num1Field.setText("");
            num2Field.setText("");

```

```

        resultLabel.setText("Calculator cleared.");
        resultLabel.setForeground(new Color(40, 70, 130));
        return;
    }
    try {
        double num1 = Double.parseDouble(num1Field.getText());
        double num2 = Double.parseDouble(num2Field.getText());
        double result = 0;
        if (e.getSource() == addButton) {
            result = num1 + num2;
            resultLabel.setText("Result = " + result);
        } else if (e.getSource() == subButton) {
            result = num1 - num2;
            resultLabel.setText("Result = " + result);
        } else if (e.getSource() == mulButton) {
            result = num1 * num2;
            resultLabel.setText("Result = " + result);
        } else if (e.getSource() == divButton) {
            if (num2 == 0) {
                resultLabel.setText("Error: Cannot divide by zero!");
                resultLabel.setForeground(Color.RED);
                return;
            }
            result = num1 / num2;
            resultLabel.setText("Result = " + result);
        }
        resultLabel.setForeground(new Color(20, 130, 70));
    } catch (NumberFormatException ex) {
        resultLabel.setText("Error: Enter valid numbers!");
        resultLabel.setForeground(Color.RED);
    }
}
}

```

OUTPUT:

Applet Viewer: CalculatorApplet.class
Applet

SIMPLE CALCULATOR Result = 25.0

First Number: 12

Second Number: 13

ADD SUBTRACT MULTIPLY
DIVIDE CLEAR

Applet Viewer: CalculatorApplet.class
Applet

SIMPLE CALCULATOR Result = 156.0

First Number: 12

Second Number: 13

ADD SUBTRACT MULTIPLY
DIVIDE CLEAR

Finished calculation of

3. Course Selection Applet

Develop a university course selection Applet where students can select:

- Department
- Semester
- Course
- Elective type

Use Choice, List, Checkbox, and Button controls wherever appropriate.

When the student clicks Register, display the selected courses and other information.

Use appropriate layout managers and event listeners.

PROGRAM:

```
import java.applet.Applet;  
import java.awt.*;  
import java.awt.event.*;
```

```
public class CourseSelectionApplet extends Applet implements ActionListener {
```

```
    Choice department, semester, course;
```

```
    Checkbox elective;
```

```
    Button register;
```

```
    TextArea result;
```

```
    public void init() {
```

```
        setLayout(new GridLayout(6, 2, 10, 10));
```

```
        setBackground(new Color(220, 235, 250));
```

```
        Label title = new Label("COURSE SELECTION", Label.CENTER);
```

```
        title.setFont(new Font("Arial", Font.BOLD, 22));
```

```
        title.setForeground(Color.BLUE);
```

```
        department = new Choice();
```

```
        department.add("CSE");
```

```
        department.add("IT");
```

```
        department.add("ECE");
```

```
        department.add("MECH");
```

```
        semester = new Choice();
```

```
        semester.add("Semester 1");
```

```
        semester.add("Semester 2");
```

```
        semester.add("Semester 3");
```

```
        semester.add("Semester 4");
```

```
        course = new Choice();
```

```
        course.add("Java");
```

```
        course.add("Data Structures");
```

```
        course.add("Database");
```

```
        course.add("Computer Networks");
```

```

elective = new Checkbox("Elective");

register = new Button("REGISTER");
register.setBackground(Color.GREEN);
result = new TextArea();
result.setFont(new Font("Arial", Font.BOLD, 16));
add(title);
add(new Label(""));
add(new Label("Department"));
add(department);
add(new Label("Semester"));
add(semester);
add(new Label("Course"));
add(course);
add(new Label("Type"));
add(elective);
add(register);
add(result);
register.addActionListener(this);
}
public void actionPerformed(ActionEvent e) {
    result.setText(
        "Registration Successful\n" +
        "Department : " + department.getSelectedItem() + "\n" +
        "Semester : " + semester.getSelectedItem() + "\n" +
        "Course : " + course.getSelectedItem() + "\n" +
        "Elective : " + elective.getState()
    );
}
}

```

OUTPUT:

University Course Registration

UNIVERSITY COURSE REGISTRATION

Department: Information Technology

Semester: Semester 2

Course: Database Management

Elective Type: ☒ Yes, Elective

REGISTER **CLEAR**

REGISTRATION DETAILS

Registration Successful!

Department : Information Technology
Semester : Semester 2
Course : Database Management
Type : Elective

University Course Registration

UNIVERSITY COURSE REGISTRATION

Department: Computer Science

Semester: Semester 5

Course: Computer Networks

Elective Type: ☐ Yes, Elective

REGISTER **CLEAR**

REGISTRATION DETAILS

Prethivika G (prethivikag5@gmail.com) is signed in

Registration Successful!

Department : Computer Science
Semester : Semester 5
Course : Computer Networks
Type : Regular

4. Traffic Signal Control Applet

Develop an AWT Applet that simulates a traffic signal with three options:

- Red
- Yellow
- Green

Use appropriate controls to allow the user to select a signal. When the user selects a signal, display the corresponding message such as STOP, WAIT, or GO.

Use appropriate event sources, event classes, and listener interfaces.

PROGRAM:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;
/*
<applet code="TrafficSignalApplet.class" width="500" height="450"></applet>
*/
public class TrafficSignalApplet extends Applet implements ItemListener {
    Checkbox red,yellow,green;
    CheckboxGroup signal;
    Label statusLabel;
    SignalCanvas display;
    int state=0;
    public void init(){
        setLayout(new BorderLayout());
        setBackground(new Color(25,27,32));
        Label title=new Label("INTERSECTION SIGNAL CONTROLLER",Label.CENTER);
        title.setFont(new Font("Segoe UI",Font.BOLD,20));
        title.setForeground(Color.WHITE);
        Panel top=new Panel(new BorderLayout());
        top.setBackground(new Color(15,17,21));
        top.add(title,BorderLayout.CENTER);
        add(top,BorderLayout.NORTH);
        signal=new CheckboxGroup();
        red=new Checkbox("RED - STOP",signal,false);
        yellow=new Checkbox("YELLOW - WAIT",signal,false);
        green=new Checkbox("GREEN - GO",signal,false);
        Font f=new Font("Segoe UI",Font.PLAIN,15);
        red.setFont(f);
        yellow.setFont(f);
        green.setFont(f);
        red.setForeground(Color.WHITE);
        yellow.setForeground(Color.WHITE);
        green.setForeground(Color.WHITE);
        Panel options=new Panel(new GridLayout(3,1,0,15));
        options.setBackground(new Color(25,27,32));
```

```

options.add(red);
options.add(yellow);
options.add(green);
add(options,BorderLayout.WEST);
display=new SignalCanvas(this);
add(display,BorderLayout.CENTER);
statusLabel=new Label("SELECT A SIGNAL",Label.CENTER);
statusLabel.setFont(new Font("Segoe UI",Font.BOLD,22));
statusLabel.setForeground(Color.LIGHT_GRAY);
add(statusLabel,BorderLayout.SOUTH);
red.addItemListener(this);
yellow.addItemListener(this);
green.addItemListener(this);
}
public void itemStateChanged(ItemEvent e){
if(red.getState()){
state=0;
statusLabel.setText("STOP");
statusLabel.setForeground(new Color(230,60,60));
}
if(yellow.getState()){
state=2;
statusLabel.setText("WAIT");
statusLabel.setForeground(new Color(230,180,40));
}
if(green.getState()){
state=1;
statusLabel.setText("GO");
statusLabel.setForeground(new Color(60,200,110));
}
display.repaint();
}
}
class SignalCanvas extends Canvas{
TrafficSignalApplet parent;
SignalCanvas(TrafficSignalApplet parent){
this.parent=parent;
setBackground(new Color(25,27,32));
}
public void paint(Graphics g){
int w=getSize().width;
int h=getSize().height;
int roadY=h-100;
g.setColor(new Color(45,48,54));

```

```

g.fillRect(0,roadY,w,100);
g.setColor(Color.WHITE);
g.fillRect(0,roadY,w,4);
for(int x=20;x<w;x+=40){
g.fillRect(x,roadY+50,20,6);
}
int poleX=w/2;
g.setColor(new Color(90,90,95));
g.fillRect(poleX-6,150,12,roadY-150);
int boxW=90;
int boxH=240;
int boxX=poleX-boxW/2;
int boxY=20;
g.setColor(new Color(35,37,42));
g.fillRoundRect(boxX,boxY,boxW,boxH,20,20);
g.setColor(new Color(60,63,70));
g.drawRoundRect(boxX,boxY,boxW,boxH,20,20);
int dia=60;
int lampX=poleX-dia/2;
int gap=(boxH-dia*3)/4;
drawLamp(g,lampX,boxY+gap,dia,parent.state==0,new Color(230,60,60),new
Color(80,40,40));
drawLamp(g,lampX,boxY+gap*2+dia,dia,parent.state==2,new
Color(230,180,40),new Color(90,75,35));
drawLamp(g,lampX,boxY+gap*3+dia*2,dia,parent.state==1,new
Color(60,200,110),new Color(35,80,50));
g.setColor(Color.WHITE);
g.drawLine(poleX-80,roadY-2,poleX+80,roadY-2);
if(parent.state==0||parent.state==2){
g.setColor(new Color(70,130,220));
g.fillRoundRect(poleX-100,roadY-34,50,26,6,6);
g.fillOval(poleX-98,roadY-12,14,14);
g.fillOval(poleX-60,roadY-12,14,14);
}
}
private void drawLamp(Graphics g,int x,int y,int dia,boolean on,Color litColor,Color
dimColor){
g.setColor(on?litColor:dimColor);
g.fillOval(x,y,dia,dia);
g.setColor(new Color(15,17,21));
g.drawOval(x,y,dia,dia);
if(on){
g.setColor(new Color(255,255,255,70));
g.fillOval(x+dia/4,y+dia/6,dia/3,dia/3);
}
}

```

```

}
}
public void update(Graphics g){
paint(g);
}
}

```

OUTPUT:



